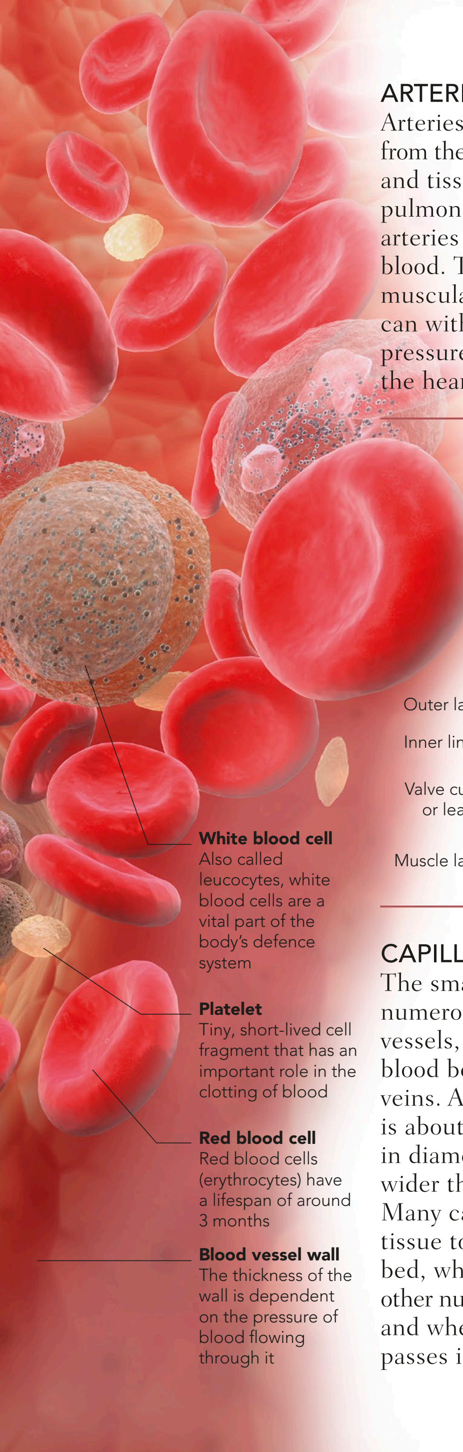
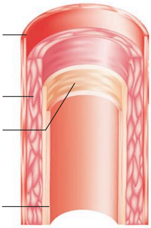


ARTERIES

Arteries carry blood away from the heart towards organs and tissues. Apart from the pulmonary arteries, all arteries carry oxygenated blood. Their thick walls and muscular and elastic layers can withstand the high pressure that occurs as the heart pumps blood.

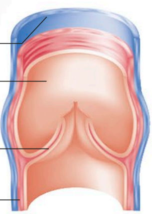


ARTERY SECTION

Four layers are found in an artery wall. The blood-carrying space, or lumen, is in the centre.

VEINS

A vein is more flexible than an artery, and its walls are thinner. The blood inside a vein is under relatively low pressure, and flows slowly and smoothly. Many larger veins, particularly the long veins in the legs, contain valves that prevent the backflow of blood, a job helped by muscles around the veins that contract during movement.



VEIN SECTION

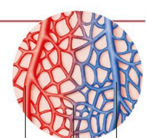
The muscle layer of a vein is thin and enclosed by two layers; the innermost layer of some veins has valves at regular intervals.

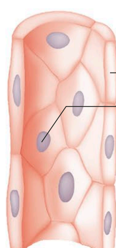
CAPILLARIES

The smallest and most numerous of the blood vessels, capillaries convey blood between arteries and veins. A typical capillary is about 0.01mm ($^1/_{2,500}$ in) in diameter, only slightly wider than a red blood cell. Many capillaries enter tissue to form a capillary bed, where oxygen and other nutrients are released, and where waste matter passes into the blood.

CAPILLARY BED

Capillaries link small arteries (arterioles) to veins (venules).





CAPILLARY WALL

The thin capillary wall allows easy movement of substances between surrounding tissues.